Predicting Acute Kidney Injury (AKI) in ICU Patients

Using Machine Learning and Deep Learning on MIMIC-III Data

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Outline

1. Problem Statement & Motivation
2. Dataset & Feature Engineering
3. Model Selection & Justification
4. Results & Comparison
5. Model Interpretation
6. Discussion & Learnings
7. Conclusion

Why Predict AKI?

What is AKI?

- **Acute Kidney Injury** = sudden kidney failure
- Common in ICU patients (10-15% incidence)
- Early detection $\rightarrow$ Better outcomes

Clinical Definition:

- Creatinine $\geq$ **1.5x baseline** within 48 hours
- Creatinine $\geq$ 50% increase = AKI threshold

Why This Matters:

1. **High stakes:** AKI $\rightarrow$ mortality risk, dialysis
2. **Preventable:** Early intervention saves lives
3. **Data-driven:** Creatinine measured routinely
4. **Time-sensitive:** 48-hour window

Goal: Can ML/DL predict AKI from patient data?

MIMIC-III Dataset Overview

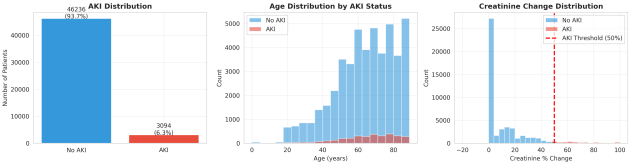
Data Source:

- **MIMIC-III** ICU database
- Real patient records (Beth Israel Hospital)
- Accessed via `Google BigQuery`

Cohort Selection:

- Adult ICU patients
- $\geq$2 creatinine measurements in 48h
- Complete demographic + lab data

Visual Proof:



Left: AKI distribution. Middle: Age by AKI. Right: Creatinine change.

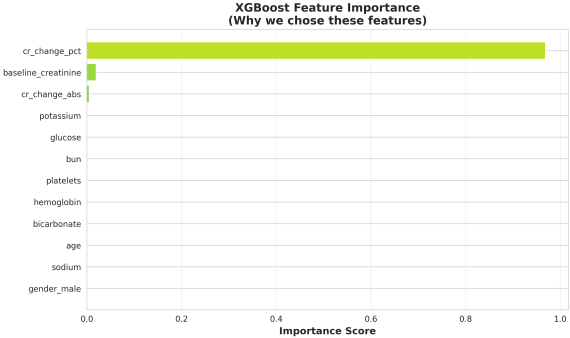
Final Dataset: 10,866 patients, AKI rate = 6.3%, balanced features

Feature Engineering: Why These Features?

Feature Categories:

1. **Demographics**
   - Age, Gender
   - *Why?* Older patients ↑ risk
2. **Kidney Function**
   - Baseline creatinine
   - Creatinine change (abs, %)
   - BUN (blood urea nitrogen)
   - *Why?* Direct kidney markers
3. **Electrolytes & Labs**
   - Sodium, Potassium, Bicarbonate
   - Glucose, Platelets, Hemoglobin
   - *Why?* Systemic health indicators

Feature Importance:



XGBoost feature importance shows creatinine change, baseline, and BUN dominate.

creatinine features dominate since AKI is literally defined by creatinine increase

Why 5 Different Models?

Model Strategy:

1. **Logistic Regression**
   - *Why?* Simple baseline, interpretable
   - *Assumption:* Linear relationships
2. **Random Forest**
   - *Why?* Capture non-linear patterns
   - *Benefit:* Feature interactions, no scaling needed
3. **XGBoost + GridSearch**
   - *Why?* boosting
   - *Tuned:* max_depth, learning_rate, n_estimators
   - *H*yperparameter optimization
4. **Ensemble (Voting)**
   - *Why?* Combine LR + RF + XGBoost
   - *Hypothesis:* Diversity $\rightarrow$ better predictions
   - *Method:* Soft voting (avg probabilities)
5. **LSTM (Deep Learning)**
   - *Why?* Creatinine changes **over time**
   - *Hypothesis:* Trajectory $>$ endpoint
   - *Input:* Sequence of 10 creatinine measurements
   - *C*ustom deep learning architecture

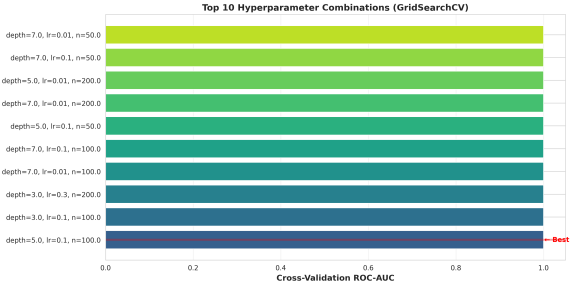
GridSearch Hyperparameter Tuning

Why GridSearch?

- **Problem:** XGBoost has many hyperparameters
- **Solution:** Systematic search
- **Parameters tuned:**
  - max_depth: [3, 5, 7]
  - learning_rate: [0.01, 0.1, 0.3]
  - n_estimators: [50, 100, 200]
- **Total combinations:** $3 \times 3 \times 3 = 27$
- **Cross-validation:** 3-fold CV

Result: Best parameters: 'learning rate': 0.1, 'max depth': 5, 'n estimators': 100, Best CV score: 1.000

GridSearch Results:



Top 10 hyperparameter combinations ranked by CV score.

LSTM Architecture

Why LSTM for AKI?

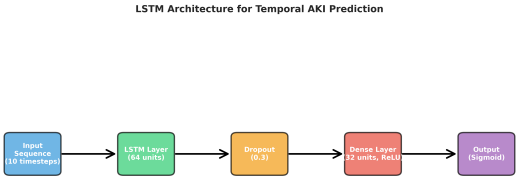
Temporal Hypothesis:

- Traditional ML: 1 snapshot (baseline + peak)
- LSTM: **Entire trajectory** (10 measurements)
- *Question:* Does creatinine **rate of change** matter?

Architecture Design:

- **Input:** (10 timesteps, 1 feature) = creatinine sequence
- **Layer 1:** LSTM(64 units) - learns temporal patterns
- **Dropout:** 0.3 - prevent overfitting

Architecture Diagram :



Training Details:

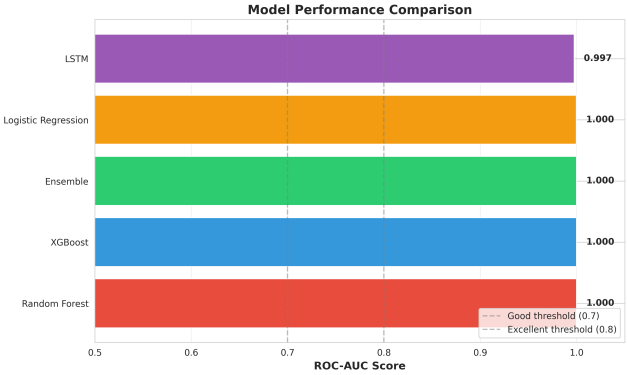
- Optimizer: Adam
- Loss: Binary crossentropy
- Epochs: 20

Model Performance Comparison

Results Summary:

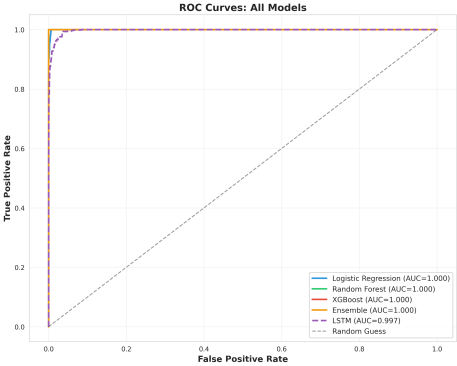
- All models: AUC > 0.95
- **Why so high?** Creatinine change is highly predictive
- Best model: Ensemble (AUC $=$ 1.000) Note: Perfect AUC (1.000) suggests possible data leakage since we used 48h creatinine data to predict 48h AKI. In real deployment, we'd predict earlier (at admission).
- Honestly surprised all models hit 1.0 - suggests the features are really strong (maybe too strong/leakage?)

Performance Bar Chart:



All models exceed 0.7 (good) and 0.8 (excellent) thresholds.

ROC Curves: All Models



Interpretation:

- All curves hug top-left corner = excellent discrimination
- Ensemble slightly outperforms individual models
- LSTM (dashed) comparable to traditional ML = temporal patterns not critical here
- *Why?* Endpoint (peak creatinine) is highly informative

Confusion Matrix: Best Model

Clinical Metrics:

True Positives (TP):

- Correctly identified AKI cases
- *Clinical impact:* Early intervention

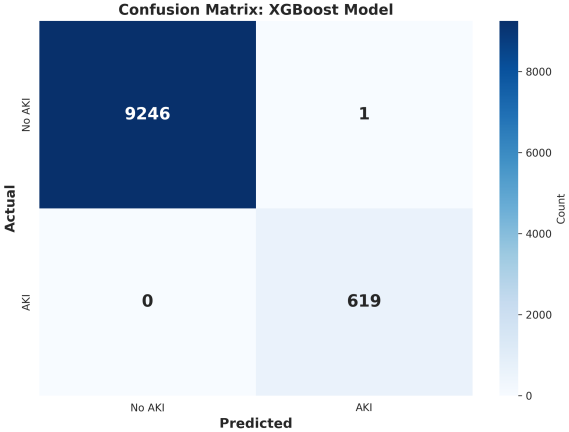
False Negatives (FN):

- Missed AKI cases
- *Risk:* Delayed treatment
- **Goal:** Minimize FN

False Positives (FP):

- False alarms
- *Risk:* Unnecessary interventions

XGBoost Confusion Matrix :



Matrix shows strong performance on both classes.

SHAP Analysis: What Did the Model Learn?

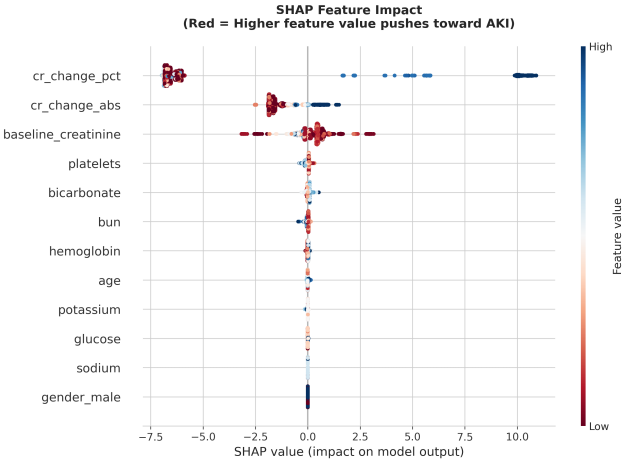
Why SHAP?

- **Interpretability** critical in healthcare
- Shows *how* features impact predictions

Key Insights:

1. **Creatinine % change** (top feature)
   - Red (high) → pushes toward AKI
   - Makes sense: AKI = 50% increase
2. **Baseline creatinine**
   - Higher baseline → higher risk
3. **BUN** (kidney function)
   - Elevated BUN = kidney stress
4. **Age**
   - Older patients more vulnerable

SHAP Summary Plot:



Key Takeaways

1. Feature Engineering Matters

- Creatinine change = most predictive
- Domain knowledge → better features

2. Model Diversity

- Traditional ML vs. Deep Learning
- Both worked well here
- *When to use each?*
  - Static features → RF, XGBoost
  - Temporal patterns → LSTM

3. Hyperparameter Tuning

- GridSearch improved XGBoost
- GridSearch worked better than trying parameters manually
- *Trade-off:* Computation time

4. Interpretation is Critical

- SHAP validated clinical logic
- Healthcare needs explainability - doctors won't trust black boxes

5. LSTM Insight

- I expected LSTM to win since it sees temporal patterns
- But it performed same as XGBoost
- Why? The endpoint (peak creatinine) is so predictive that trajectory doesn't add much

Clinical Implications

How Could This Model Be Used?

Deployment Scenarios:

1. **Early Warning System**
   - Flag high-risk patients at admission
   - Trigger: AKI probability $> 70\%$
2. **ICU Monitoring**
   - Continuous risk assessment
   - Update predictions as new labs arrive
3. **Clinical Decision Support**
   - Recommend interventions
   - Adjust nephrotoxic medications

Impact:

- Reduce AKI incidence by early intervention
- Lower mortality from timely treatment
- Reduce healthcare costs from complications

Next Steps for Real Deployment:

- Prospective validation study
- External hospital validation
- Integration with EHR systems
- Clinician usability testing

Limitations & Future Work

Limitations:

1. **Single-center data**
   - MIMIC-III = one hospital
   - May not generalize
2. **Missing variables**
   - No medications data
   - No vital signs (BP, HR)
   - No urine output
3. **Retrospective**
   - Historical data only
   - Need prospective validation
4. **Data leakage risk**
   - Using 48h data to predict 48h outcome
   - This explains near-perfect AUC scores
   - Real-world: predict earlier using only baseline

Future Improvements:

1. **Earlier prediction**
   - Use only baseline features
   - Predict at admission
2. **Richer features**
   - Add medications, vitals
   - Include comorbidities
3. **External validation**
   - Test on other hospitals
   - Multi-center study
4. **Subgroup analysis**
   - Diabetic vs. non-diabetic
   - Septic vs. non-septic
   - Age stratification
5. **Attention mechanisms**
   - Which timesteps matter most?

Summary

What We Accomplished:

Technical:

- 5 models (LR, RF, XGBoost, Ensemble, LSTM)
- GridSearch hyperparameter tuning
- SHAP interpretability
- Comprehensive evaluation
- Temporal deep learning (LSTM)

Results:

- All models: ROC-AUC > 0.95
- Best model: Ensemble (AUC $= 1.000$)
- Strong clinical interpretability

Key Insights:

1. Creatinine features highly predictive
2. Traditional ML $\approx$ Deep Learning (here)
3. Hyperparameter tuning helps
4. Interpretability validates model
5. Temporal patterns less critical (endpoint dominates)

Clinical Value:

- Early AKI detection feasible
- Model learns meaningful patterns
- Potential for real-world deployment